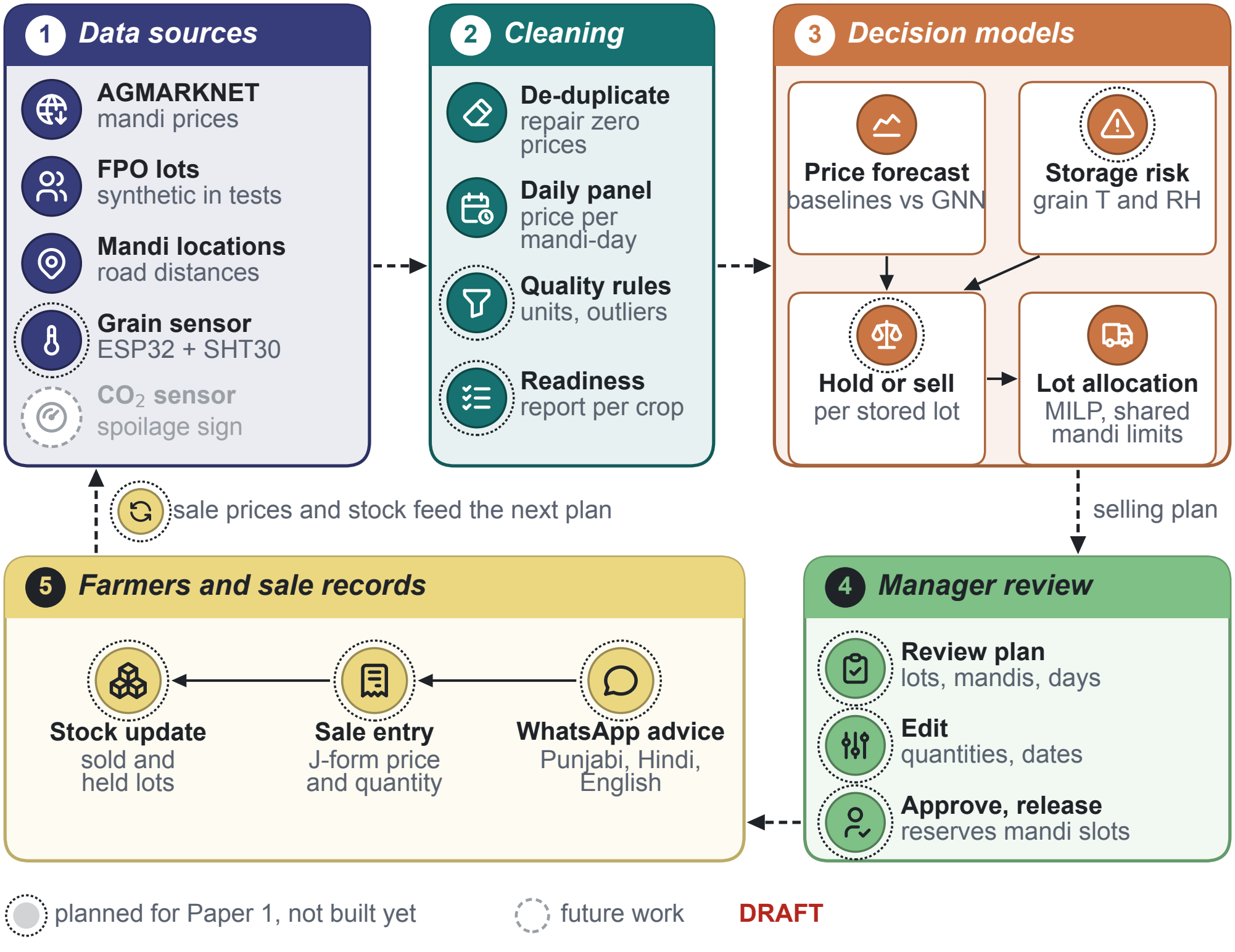

1 Data sources
AGMARKNET
mandi prices
FPO lots
synthetic in tests
Mandi locations
road distances
Grain sensor
ESP32 + SHT30
CO2 sensor
spoilage sign
2 Cleaning
De-duplicate
repair zero prices
Daily panel
price per mandi-day
Quality rules
units, outliers
Readiness
report per crop
3 Decision models
Price forecast
baselines vs GNN
Storage risk
grain T and RH
Hold or sell
per stored lot
Lot allocation
MILP, shared mandi limits
selling plan
4 Manager review
Review plan
lots, mandis, days
Edit
quantities, dates
Approve, release
reserves mandi slots
5 Farmers and sale records
Stock update
sold and held lots
Sale entry
J-form price and quantity
WhatsApp advice
Punjabi, Hindi, English
sale prices and stock feed the next plan
planned for Paper 1, not built yet
future work
DRAFT